RAJALAKSHMI ENGINEERING COLLEGE

Priyadharshini A 251001198-2025-IT

P2

| | |
|---|---|
| **Started on** | Wednesday, 30 September 2026, 9:24 PM |
| **State** | Finished |
| **Completed on** | Wednesday, 30 September 2026, 9:41 PM |
| **Time taken** | 16 mins 26 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

**For example:**

| Input | Result |
|---|---|
| 1<br>3 10 17 57<br>6<br>2 7 10 15 57 246 | 10 57 |

**Answer:** (penalty regime: 0 %)

```
#include <stdio.h>
#include <stdlib.h>

int main()
{
    int t, n1, n2, i, j, first;

    scanf("%d", &t);

    while(t--)
    {
        scanf("%d", &n1);

        int *a = malloc(n1 * sizeof(int));
```

```
15
16          for(i = 0; i < n1; i++)
17              scanf("%d", &a[i]);
18
19          scanf("%d", &n2);
20
21          int *b = malloc(n2 * sizeof(int));
22
23          for(i = 0; i < n2; i++)
24              scanf("%d", &b[i]);
25
26          i = 0;
27          j = 0;
28          first = 1;
29
30          while(i < n1 && j < n2)
31          {
32              if(a[i] == b[j])
33              {
34                  if(first == 0)
35                      printf(" ");
36
37                  printf("%d", a[i]);
38                  first = 0;
39
40                  while(i < n1 && a[i] == a[i + 1] && i + 1 < n1)
41                      i++;
42
43                  while(j < n2 && b[j] == b[j + 1] && j + 1 < n2)
44                      j++;
45
46                  i++;
47                  j++;
48              }
49              else if(a[i] < b[j])
50              {
51                  i++;
52              }
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 1<br>3 10 17 57<br>6<br>2 7 10 15 57 246 | 10 57 | 10 57 | ✔ |
| ✔ | 1<br>6 1 2 3 4 5 6<br>2<br>1 6 | 1 6 | 1 6 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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